23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY
PHASE – 2 (Design)
Architectural Modelling (CO2)

Group No : 14

Team Members :

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1. Problem Title : Attendance Management System

STRUCTURAL DESIGN

2. Class diagram

```
┌──────────────────────────────────────────────────────────────────┐
│                    <<abstract>>  User                            │
├──────────────────────────────────────────────────────────────────┤
│ - userId  : String                                               │
│ - name    : String                                               │
│ - email   : String                                               │
│ - role    : String                                               │
├──────────────────────────────────────────────────────────────────┤
│ + displayDashboard(manager: AttendanceManager) <<abstract>>      │
│ + getters / setters                                              │
└──────────────────────┬──────────────────────┬────────────────────┘
                       │ extends              │ extends
            ┌──────────┴──────────┐  ┌────────┴────────┐
            │      Student        │  │     Teacher     │
            ├─────────────────────┤  ├─────────────────┤
            │ - program           │  │ - department    │
            │ - semester          │  │ - employeeCode  │
            ├─────────────────────┤  ├─────────────────┤
            │ + displayDashboard  │  │ + displayDashboard│
            └─────────────────────┘  └─────────────────┘


┌──────────────────┐     ┌──────────────────────────────────┐
│    Subject       │     │         AttendanceRecord         │
├──────────────────┤     ├──────────────────────────────────┤
│ - subjectId      │     │ - recordId    : String           │
│ - subjectName    │     │ - studentId   : String           │
│ - subjectCode    │     │ - subjectId   : String           │
│ - teacherId      │     │ - date        : LocalDate        │
│ - totalClasses   │     │ - status      : AttendanceStatus │
├──────────────────┤     │ - remarks     : String           │
│ + getters/setters│     │ - recordedBy  : String           │
│ + incrementTotal │     ├──────────────────────────────────┤
└──────────────────┘     │ + isPresent() / isAbsent()       │
                         │ + setStatus() [controlled edit]  │
                         └──────────────────────────────────┘
```

```
<<enum>> AttendanceStatus { PRESENT, ABSENT, LEAVE }

┌──────────────────────────────────────────────────────────────┐
│                   AttendanceSummary (value object)           │
├──────────────────────────────────────────────────────────────┤
│ - studentId / studentName / subjectId / subjectName          │
│ - totalClasses / classesAttended / leavesApproved            │
│ - attendancePercent : double  (calculated on construction)   │
│ - allowableAbsences : int     (calculated on construction)   │
│ - threshold         : double                                 │
├──────────────────────────────────────────────────────────────┤
│ + isCompliant() : boolean                                    │
│ + getStatusLabel() : String                                  │
└──────────────────────────────────────────────────────────────┘


┌──────────────────────────────────────────────────────────────┐
│                   AttendanceManager (service)                │
├──────────────────────────────────────────────────────────────┤
│ - students   : Map<String, Student>                          │
│ - teachers   : Map<String, Teacher>                          │
│ - subjects   : Map<String, Subject>                          │
│ - records    : List<AttendanceRecord>                        │
│ - enrollment : Map<String, Set<String>>                      │
│ - minAttendanceThreshold : double                            │
├──────────────────────────────────────────────────────────────┤
│ + addStudent / addTeacher / addSubject / enrollStudent       │
│ + recordAttendance()                                         │
│ + recordClassSession()   [bulk — whole class at once]        │
│ + editAttendance()                                           │
│ + computeSummary()       [core calculation]                  │
│ + printStudentSubjectReport()                                │
│ + printSubjectAttendanceSummary()                            │
│ + printLowAttendanceReport()                                 │
│ + printAttendanceLog()   [audit trail]                       │
│ + getSubjectsForStudent() / getSubjectsForTeacher()          │
└──────────────────────────────────────────────────────────────┘
```

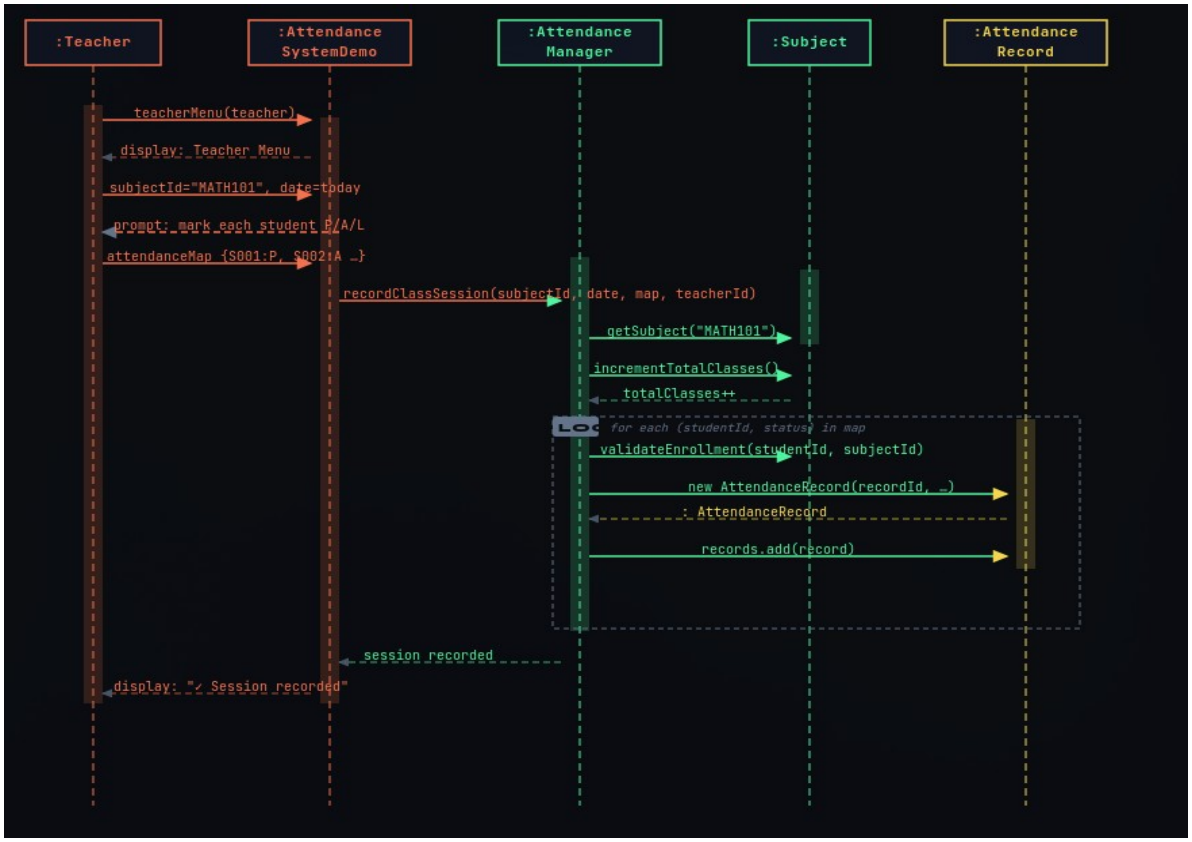
BEHAVIUORAL DESIGN

3. Use-case diagram
    a. List of actors
        1. Student
        2. Faculty
        3. Admin
    b. List of use-cases
        1. Login :

Users (both students and faculty) log into the system using credentials.

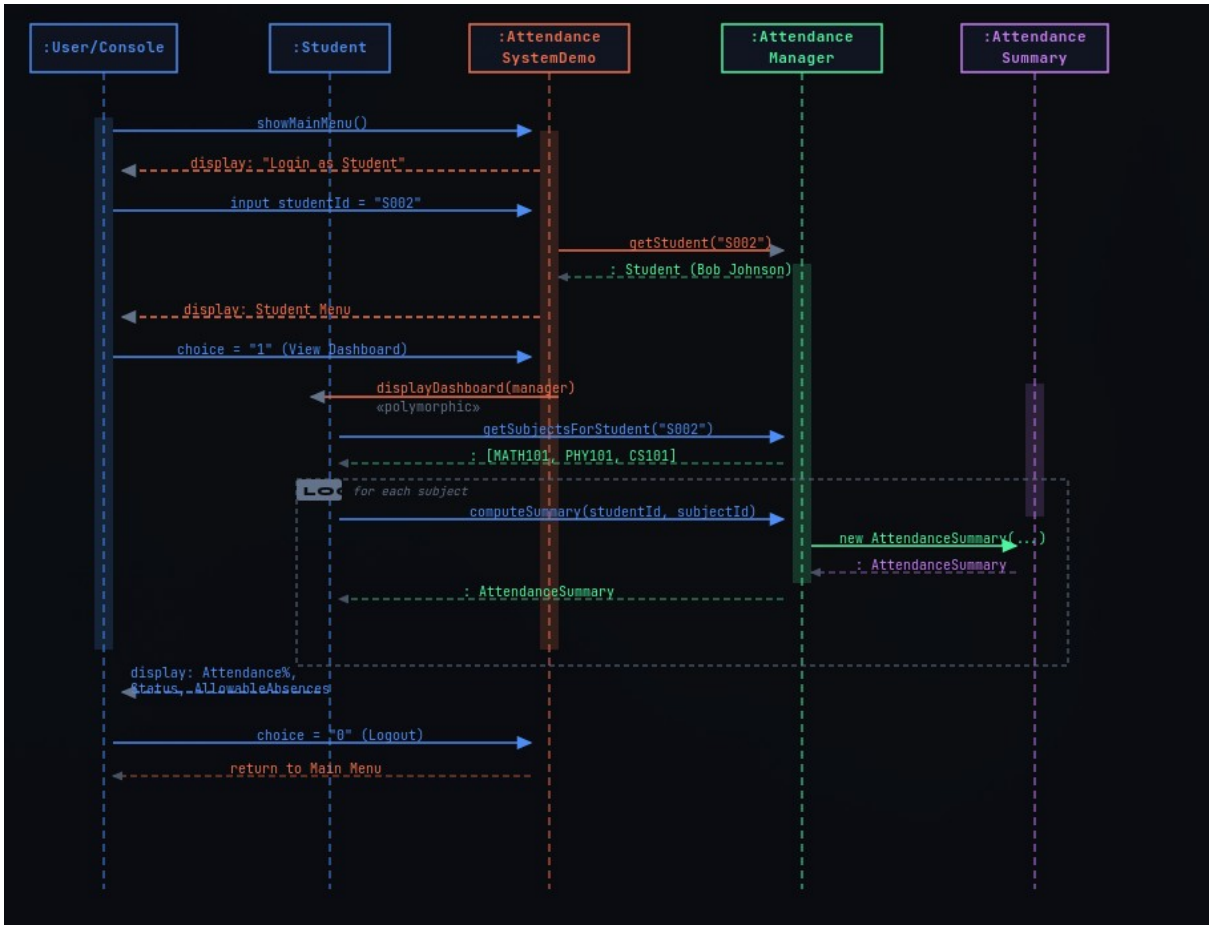
2. View Attendance :
   Students can check their attendance records.
3. Mark Attendance :
   Faculty marks attendance for students
4. Update Attendance :
   Faculty edits attendance if needed.
5. Manage Users :
   Admin adds/removes students and faculty.
6. Assign Courses :
   Admin assigns courses to faculty and students.

4. Sequence diagram (use UML notations)
   a. Teacher Records Attendance



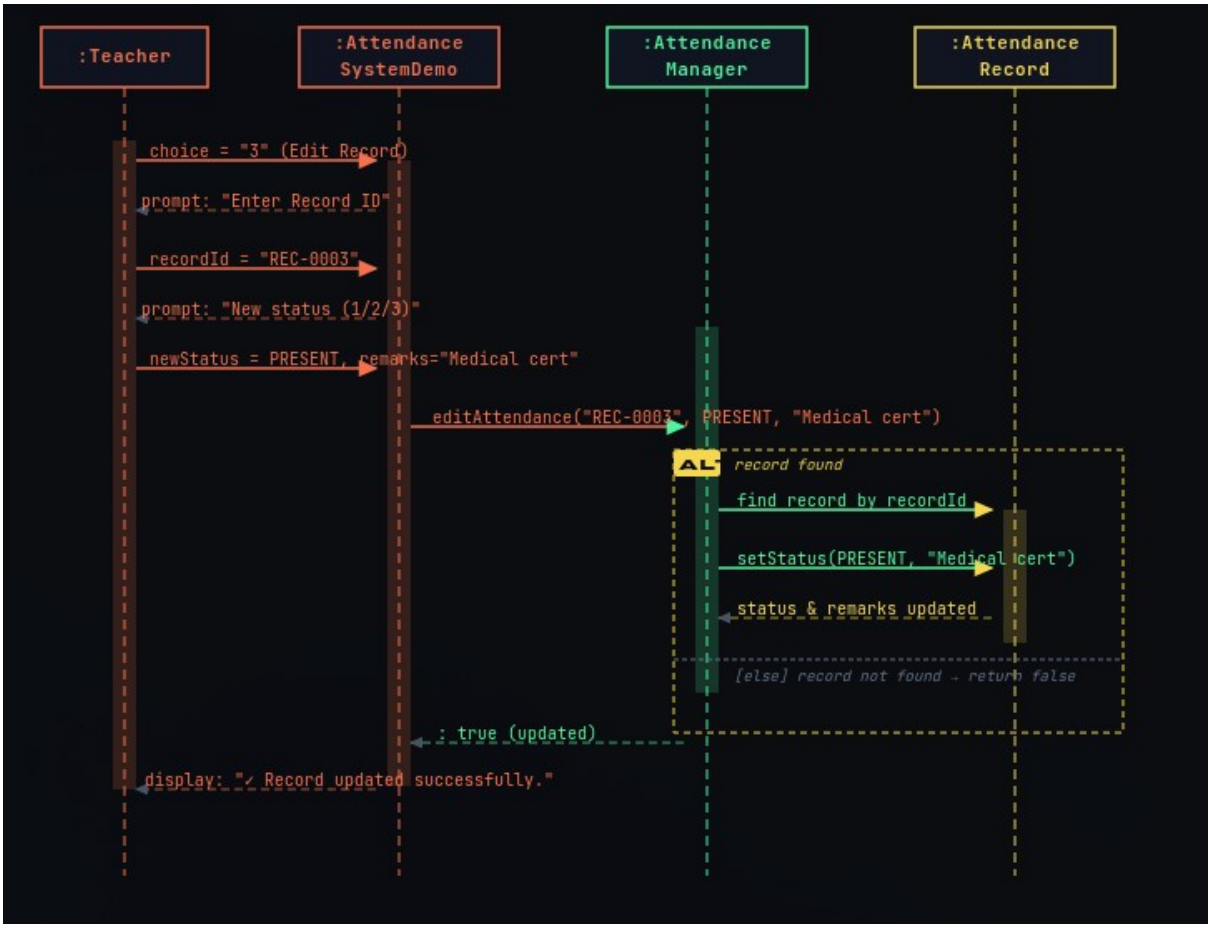
Scenario: The teacher selects a subject (e.g., MATH101) and inputs the attendance status for each enrolled student. The system validates the enrollment and creates new attendance records for the session.

b. Student Login & View Dashboard



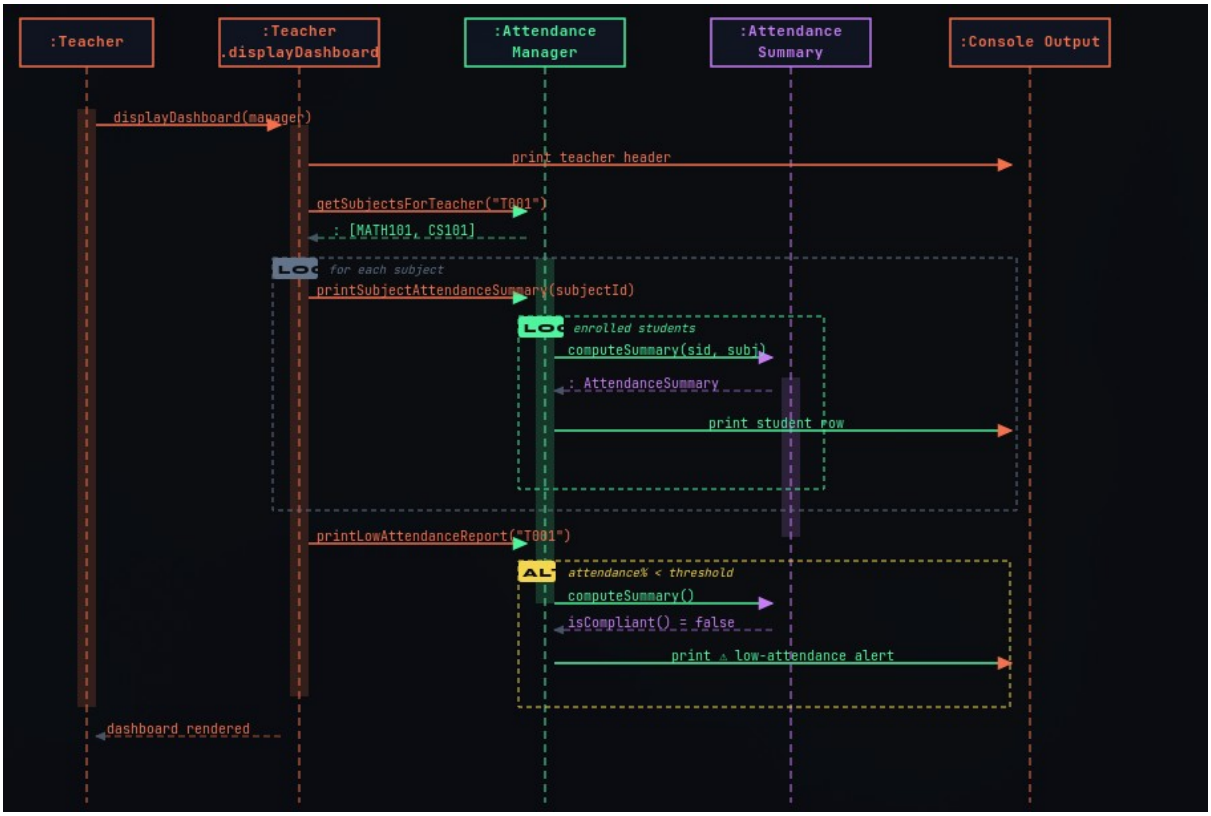
Scenario: A student logs into the console. The system fetches their enrolled subjects and computes an attendance summary for each, displaying their status and allowable absences.

c. Edit Attendance Record



Scenario: A teacher identifies an error in an existing attendance entry. They enter the specific Record ID and provide the updated status (e.g., changing from Absent to Present due to a medical certificate).

c. Teacher Dashboard & Alerts



Scenario: The teacher requests a dashboard view. The system iterates through the teacher's subjects, generates summaries for all enrolled students, and prints specific alerts for students whose attendance falls below the required threshold.